14

Multi-segment configuration guidelines for half-duplex Ethernet systems

Objectives

This chapter provides some design and evaluation insights into multi-segment configuration guidelines for half-duplex Ethernet systems. Study of this chapter will provide:

- An understanding of approaches for verifying the configuration of a half-duplex shared Ethernet channels
- Information on Model I and Model II guidelines laid down in the IEEE 802.3 standard
- Rules for combining multiple segments with repeater hubs
- Detailed analysis of methods of building complex half-duplex Ethernet systems operating at 10, 100, and, 1000 Mbps
- Examples of sample network configurations

Note: This topic is only relevant for CSMA/CD systems. Most modern Ethernet systems are full duplex switched systems, with no timing constraints.

14.1 Introduction

The basic configuration of simple half-duplex Ethernet systems using a single medium is easily done by studying the properties of the medium to be used and studying the standard rules. But, when it comes to more complex half-duplex systems based on repeater hubs, multi-segment configuration guidelines need to be studied and applied.

The official configuration guidelines lay down two methods for configuring these systems, the methods being called Model I and Model II.

Model I comprises of a set of ready-to-use rules. Model II lays down calculation aids for evaluation of more complex topologies that cannot be easily configured by the application of Model I rules.

The guidelines are applicable to equipment described in, or made to confirm to, the IEEE 802.3 standard. The segments must be built as per recommendations of the standard. If this is not adhered to, verification and evaluation of operations in terms of signal timing is not possible.

Proper documentation of each network link must be prepared when it is installed. Information about the cable length of each segment, cable types, cable ID numbers etc, should be recorded in the documentation. The IEEE standard recommends creation of documentation formats based on Table 14.1 shown below:

	Horrizontal Cabling	Transceiver Cabling	Wiring Closet Patch Cords	Station Patch Cords
Length				
Type				
Cable Manufacturer				
Cable Code/ID				

Table 14.1
Cable segment documentation form

14.2 Defining collision domains

The multi-segment configuration guidelines apply to the MAC protocol and half-duplex Ethernet collision domains described earlier in this manual.

A collision domain is defined as a network within which there will be a collision if two computers attached to the system transmit at the same time. An Ethernet system made up of a single segment or of a multiple segments linked together with repeater hubs make a single collision domain. A typical collision domain is shown in Figure 14.1.

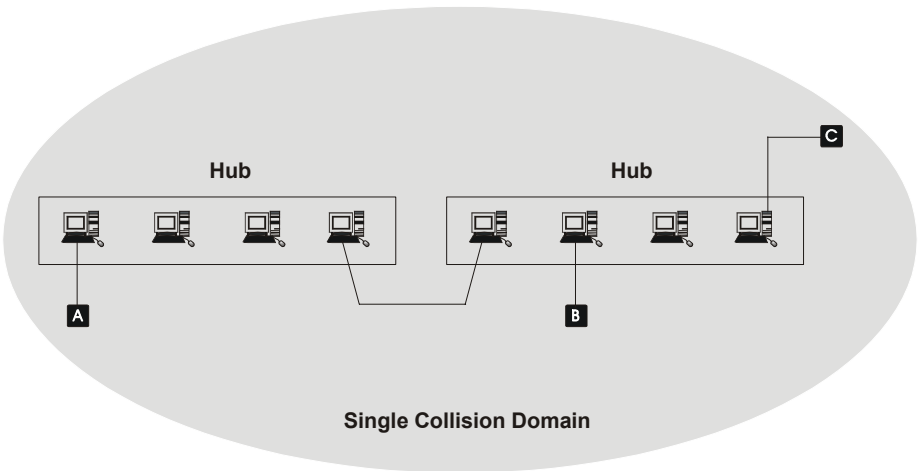


Figure 14.1
Repeaters create a single collision domain

All segments within a collision domain must operate at the same speed. For this reason, there are separate configuration guidelines for Ethernet segments with of different speeds.

IEEE 802.3 lays down guidelines for the operation of a single half-duplex LAN, and does not say anything about combining multiple single collision domains. Switching hubs, however, enable the creation of a new collision domain on each port of a switching hub, thereby linking many networks together. Segments with different speeds can also be linked this way.

14.3 Model I configuration guidelines for 10 Mbps systems

Model I in the IEEE 802.3 standard describes a set of multi-segment configuration rules for combining various 10 Mbps Ethernet segments. Most of the terms and phrases used below have been taken directly from the IEEE standard.

The guidelines are as follows:

Repeater sets are required for all segment interconnections. A 'repeater set' is a repeater and its associated transceivers if any. Repeaters must comply with all specifications in the IEEE 802.3 standard.

MAUs that are a part of repeater sets do not count towards the maximum number of MAUs on a segment. Twisted-pair, fiber optic and thin coax repeater hubs typically use internal MAUs located inside each port of the repeater. Thick Ethernet repeaters use an outboard MAU to connect to the thick coax.

The transmission path permitted between any two DTEs may consist of up to five segments, four repeater sets (including optional AUIs), two MAUs, and two AUIs. The repeater sets are assumed to have their own MAUs, which are not counted in this rule.

AUI cables for 10BaseFP and 10BaseFL shall not exceed 25 m. Since two MAUs per segment are required, 25 m per MAU results in a total AUI cable length of 50 m per segment.

When a transmission path consists of four repeaters and five segments, up to three of the segments may be mixing segments and the remainder must be link segments. When five segments are present, each fiber optic link segment (FOIRL, 10BaseFB, or 10BaseFL) shall not exceed 500 m, and each 10BaseFP segment shall not exceed 300 m. A mixing segment is one that may have more than two medium dependent interfaces attached to it, e.g. a coaxial cable segment. A link segment is a point-to-point full-duplex medium that connects two and only two MAUs.

When a transmission path consists of three repeater sets and four segments, the following restrictions apply:

The maximum allowable length of any inter-repeater fiber segment shall not exceed 1000 m for FOIRL, 10BaseFB, and 10BaseFL segments and shall not exceed 700 m for 10BaseFP segments.

The maximum allowable length of any repeater to DTE fiber segment shall not exceed 400 m for 10BaseFL segments and shall not exceed 300 m for 10BaseFP segments and 400 m for segments terminated in a 10BaseFL MAU.

There is no restriction on the number of mixing segments in this case. When using three repeater sets and four segments, all segments may be mixing segments if so desired.

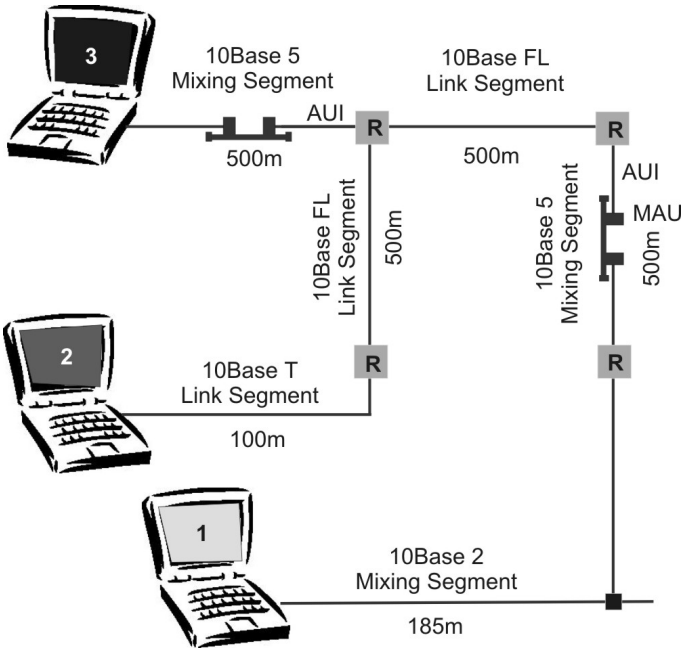


Figure 14.2
Model I 10 Mbps configuration

Figure 14.2 shows an example of one possible maximum Ethernet configuration that meets the Model I configuration rules. The maximum packet transmission path in this system is between station 1 and station 2, since there are four repeaters and five media segments in that particular path. Two of the segments in the path are mixing segments, and the other three are link segments.

The Model I configuration rules are based on conservative timing calculations. That however should not be taken to mean that these rules could be relaxed. In spite of the allowances made in the standards for manufacturing tolerances and equipment variances, there isn't much margin left in maximum-sized Ethernet networks. For maximum performance and reliability, it is better to conform to the published guidelines.

14.4 Model II configuration guidelines for 10 Mbps

The Model II configuration guidelines provide a set of calculation aids that make it possible to check the validity of more complex Ethernet systems. This is a simple process based on multiplication and addition.

There are two sets of calculations provided in the standard that must be performed for each Ethernet system. The first set of calculations verifies the round-trip signal delay time, while the second set verifies that the amount of inter-frame gap shrinkage is within normal limits. Both calculations are based on network models that evaluate the worst-case path through the network.

14.4.1 Models of networks and transmission delay values

The network models and transmission delay values provided in the Model 2 guidelines deliberately hide a lot of complexity while still making it possible to calculate the timing values for any Ethernet system. Each component in an Ethernet system provides a certain amount of transmission delay and all of these are listed in the 802.3 standard in detail.

The choice of equipment used influences the transmission and delay of Ethernet signals through the system.

Complex delay calculations and delay considerations covered earlier are found in the Model II guidelines. The standard also provides a set of network models and segment delay values. The worst-case path of a system is defined as the path through a network that has the longest segments and the most number of repeaters between any two stations. A standard network model used in calculating the round-trip timing of a system's worst-case path is shown in Figure 14.3 This calculation model, which is perhaps the most commonly used one, includes a left and right end segment, and many middle segments. The number of middle segments used in the calculation is dependent on individual systems, although a minimum number is shown in the Figure 14.3.

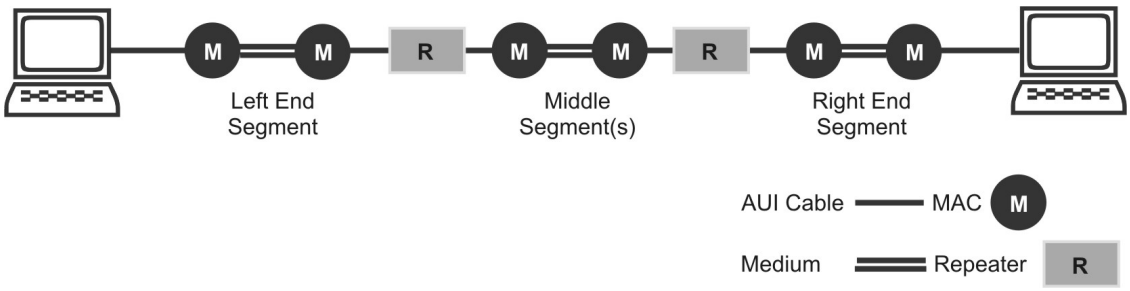


Figure 14.3
Network model for round-trip timing

A similar model is used to check the worst-case path's round-trip timing on any network under evaluation. Later we will use this model to evaluate the round-trip timing of two sample networks Interframe gap shrinkage is also calculated by using a similar model, as will be demonstrated later.

14.4.2 The worst-case path

The first step is to locate the path with the maximum delay in a network. As defined earlier, this is the path with the longest round-trip time and the largest number of repeaters between two stations. In some cases, there may be more than one worst-case path in the system. If such a problem is encountered then it is prudent to identify all the paths through the given network and classify them using the definition of a worst-case path. Once this is done, calculate the round-trip timing or interframe gap for each path. Whenever the results exceed the limits prescribed by the standard classify the network as 'failed to pass the test'.

A complete and up-to-date map of the network should be available to find the worst-case path between two stations. The information needed in such a map must include:

- Segment types used (twisted pair, fiber optic, coax)
- Segment length
- Repeater locations for the entire system
- Segment and repeater layouts for the system

Once the worst-case path is found, the next thing needed is to draw your path based on the standard model shown in Figure 14.3. This is done by assigning the segment at one end of the worst-case path to be a left end segment, leaving a right end segment with one or more middle segments.

For doing this, draw a sketch of your worst-case path, noting the segment types and lengths. Then arbitrarily assign one of the end segments to be the left end; the type of

segment assigned doesn't really matter. This leaves a right end segment. Consequently all other segments in the worst-case path are classified as middle segments.

14.4.3 Calculating the round-trip delay time

If and when any two stations on a half-duplex Ethernet channel transmit at the same time, they must have fair access to the system. This is one of the issues that the configuration guidelines try to address. When this is achieved, each station attempting to transmit must be notified of channel contention (a possible collision). Each station receiving a collision signal within the correct collision-timing window achieves this.

Calculating the total path delay, or *round-trip timing*, of the worst-case path determines whether an Ethernet system meets the limits or not. This is done using segment delay values. Each Ethernet media type has a value provided in terms of bit time that eventually determines the delay value. A bit time is the amount of time required to send one data bit on the network. For a 10 Mbps Ethernet system the value is 100 nanoseconds (ns). Table 14.2 gives the segment delay values provided in the standard. These are used in calculating the total worst-case path delay.

Segment Type	Max Length (meters)	Left End		Middle Segment		Right End		RT Delay/meter
		Base	Max	Base	Max	Base	Max	
10Base5	500	11.75	55.05	46.5	89.8	169.5	212.8	0.0866
10Base2	185	11.75	30.73	146.5	65.48	169.5	188.48	0.1026
10BaseT	100	15.25	26.55	42.0	53.3	165	176.3	0.113
10BaseFL	2000	12.25	212.25	33.5	233.5	156.5	356.5	0.1
Excess AUI	48	0	4.88	0	4.88	0	4.88	0.1026

Table 14.2
Round trip delay value in bit times

The total round-trip delay is found by adding up the delay values found on the worst-case path in network. Once calculations have been done on the segment delay values for each segment in the worst-case path, add the segment delay values together to find the total path delay. The standard recommends addition of a margin of 5 bit times to this total. If the result is less than or equal to 575 bit times, the path passes the test.

This value ensures that a station at the end of a worst-case path will be notified of a collision and stop transmitting within 575 bit times. This includes 511 bits of the frame plus the 64 bits of frame preamble and start frame delimiter (511 + 64 = 575). Once it is known that the round-trip timing for the worst-case path is okay, and then one can be sure that all other paths must be okay as well.

There is need to check one more item in the calculation for total path delay. If the path being checked has left and right end segments of different segment types, then check the path twice. The first time through, use the left end path delay values of one of the segment types, and the second time through use the left end path delay values of the other segment type. The total path delay must pass the delay calculations no matter which set of path delay values are used.

14.4.4 The inter-frame gap shrinkage

The inter-frame gap is a 96-bit time delay provided between frame transmissions to allow the network interfaces and other components some recovery time between frames. As

frames travel through an Ethernet system, the variable timing delays in network components combined with the effects of signal reconstruction circuits in the repeaters can result in an apparent shrinkage of the inter-frame gap.

Too small a gap between frames can overrun the frame reception capability of network interfaces, leading to lost frames. Therefore, it's important to ensure that a minimum inter-frame gap is maintained at all receivers (stations).

The network model for checking the inter-frame gap shrinkage is shown in figure 14.4.

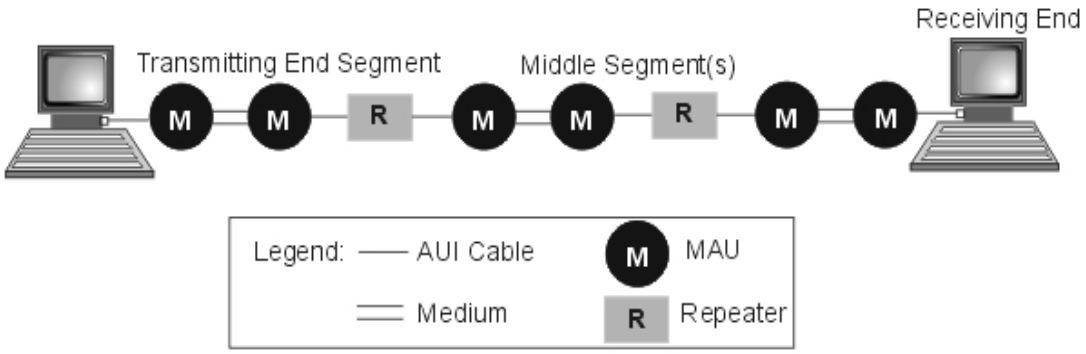


Figure 14.4
Network model for interframe gap shrinkage

Figure 14.4 looks a lot like the round-trip path delay model (Figure 14.3), except that it includes a *transmitting end* segment. When one is doing the calculations for inter-frame gap shrinkage, only the transmitting end and the middle segments are of interest, since only signals on these segments must travel through a repeater to reach the receiving end station. The final segment connected to the receiving end station does not contribute any gap shrinkage and is therefore not included in the interframe gap calculations. Table 14.3 gives the values used for calculating inter-frame gap shrinkage.

Segment Type	Transmitting End	Mid-Segment
Coax	16	11
Link Segment	10.5	8

Table 14.3
Interframe gap shrinkage in bit times

When the receive and transmit end segments are not of the same media type, the standard lays down use of the end segment with the largest number of shrinkage bit times as the transmitting end for the purposes of this calculation. This will provide the worst-case value for interframe gap shrinkage. If the total is less than or equal to 49 bit times, then the worst-case path passes the shrinkage test.

14.5 Model 1-configuration guidelines for Fast Ethernet

Transmission system Model 1 of the Fast Ethernet standard ensures that the important Fast Ethernet timing requirements are met, so that the medium access control (MAC) protocol will function correctly.

The basic rules for Fast Ethernet configuration include:

- All copper (twisted-pair) segments must be less than or equal to 100 meters in length
- Fiber segments must be less than or equal to 412 meters in length

- If Medium Independent Interface (MII) cables are used, they must not exceed 0.5 meters each

When it comes to evaluating network timing, delays attributable to the MII do not need to be accounted for separately, since these delays are incorporated into station and repeater delays.

Table 14.4 shows the maximum collision domain diameter for segments using Class I and Class II repeaters. The maximum collision domain diameter in a given Fast Ethernet system is the longest distance between any two stations (DTEs) in the collision domain.

Repeater Type	All Copper	All Fiber	Copper+Fiber eg.T4+FX	Copeer+Fiber eg. TX+FX
DTE-DTE Single Segment	100	412	N/A	N/A
One Class I Repeater	200	272	231(1)	260.8 (1)
One Class II Repeater	200	320	N/A (2)	308.8 (1)
Two Class II Repeaters	205	228	N/A (2)	216.2 (3)

Notes:(1) Assumes 100 m. Of twisted-pair cabling and one fiber link
(2) FX Fiber and T4 copper cannot be linked in a class II repeater
(3) Assumes 105 m. Of twisted-pair cable and one fiber link

Table 14.4
Maximum Fast Ethernet collision domain in meters as per Model I guidelines

The first row in the above table shows that a DTE-to-DTE (station-to-station) link with no intervening repeater may be made up of a maximum of 100 meters of copper, or 412 meters of fiber optic cable. The next row provides the maximum collision domain diameter when using a Class I repeater, including the case of all-twisted-pair and all-fiber optic cables, or a network with a mix of twisted-pair and fiber cables. The third row shows the maximum collision domain length with a single Class II repeater in the link.

The last row of the shows the maximum collision domain allowed when two Class II repeaters are used in a link. In this last configuration, the total twisted-pair segment length is assumed 105 meters on the mixed fiber and twisted-pair segment. This includes 100 meters for the segment length from the repeater port to the station, and five meters for a short segment that links the two repeaters together in a wiring closet.

Figure 14.5 shows an example of a maximum configuration based on the 100 Mbps simplified guidelines we've just seen. Note that the maximum collision domain diameter includes the distance:

A (100 m) + B (5 m) + C (100 m)

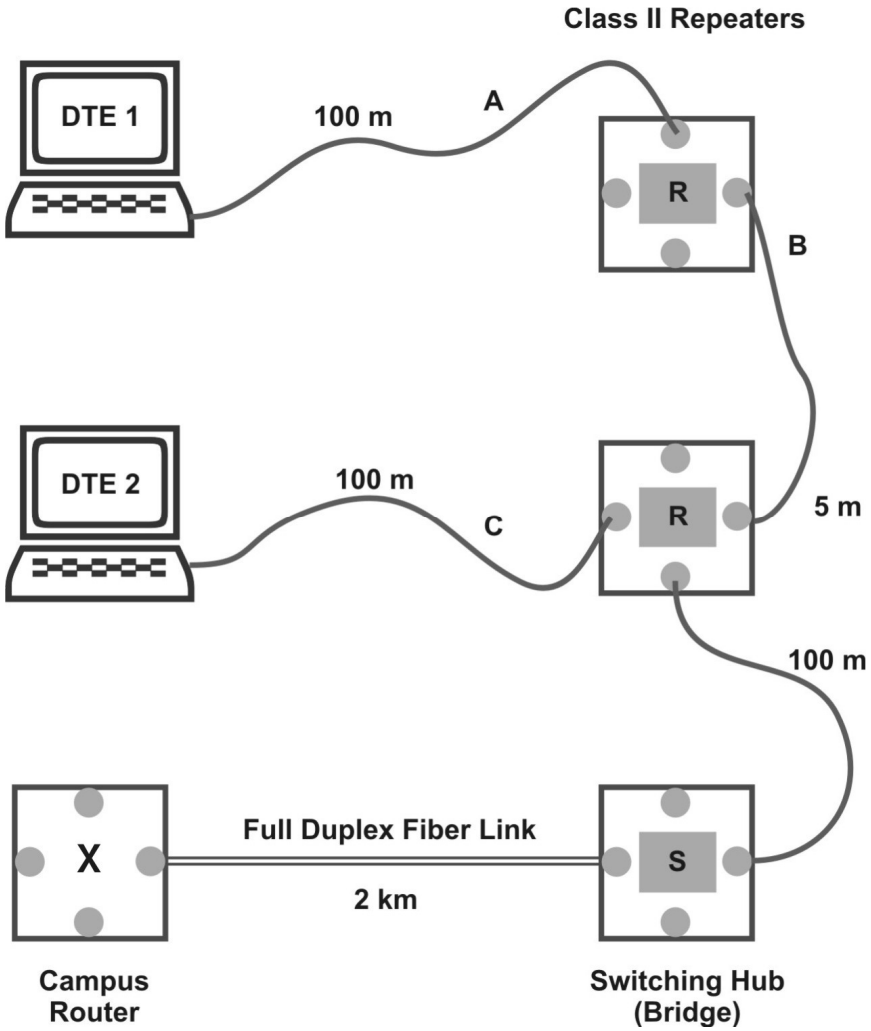


Figure 14.5
One possible maximum 100 Mbps configuration

The inter-repeater segment length can be longer than 5 m as long as the maximum diameter of the collision domain does not exceed the guidelines for the segment types and repeaters being used. Segment *B* in above figure could be 10 meters in length, for instance, as long as other segment lengths are adjusted to keep the maximum collision diameter to 205 meters. While it's possible to vary the length of the inter-repeater segment in this fashion, you should be wary of doing so and carefully consider the consequences.

14.5.1 Longer inter-repeater links

Using longer inter-repeater links has some shortcomings. Their use makes network timing rely on the use of shorter than standard segments from the repeater ports to the stations, which could cause confusion and problems later on. These days it assumed that twisted-pair segment lengths could be up to 100 meters long. Because of that, a new segment that's 100 meters long could be attached to a system with a long inter-repeater link later. In this case, the maximum diameter between some stations could then become 210 meters. If the signal delay on this long path exceeds 512 bit times, then the network may experience problems, such as late collisions. This can be avoided by keeping the length of inter-repeater segments to five meters or less.

A switching hub is just another station (DTE) as far as the guidelines for a collision domain are concerned. The switching hub shown in figure 14.5 provides a way to link separate network technologies – in this case, a standard 100BaseT segment and a full-duplex Ethernet link. The switching hub is shown linked to a campus router with a full-duplex fiber link that spans up to two kilometers. This makes it possible to provide a 100 Mbps Ethernet connection to the rest of a campus network using a router port located in a central section of the network.

Figure 14.6 shows an example of a maximum configuration based on a mixture of fiber optic and copper segments. Note that there are two paths representing the maximum collision domain diameter. This includes the distance A (100 m) + C (208.8 m), or the distance B (100 m) + C (208.8 m), for a total of 308.8 meters in both cases.

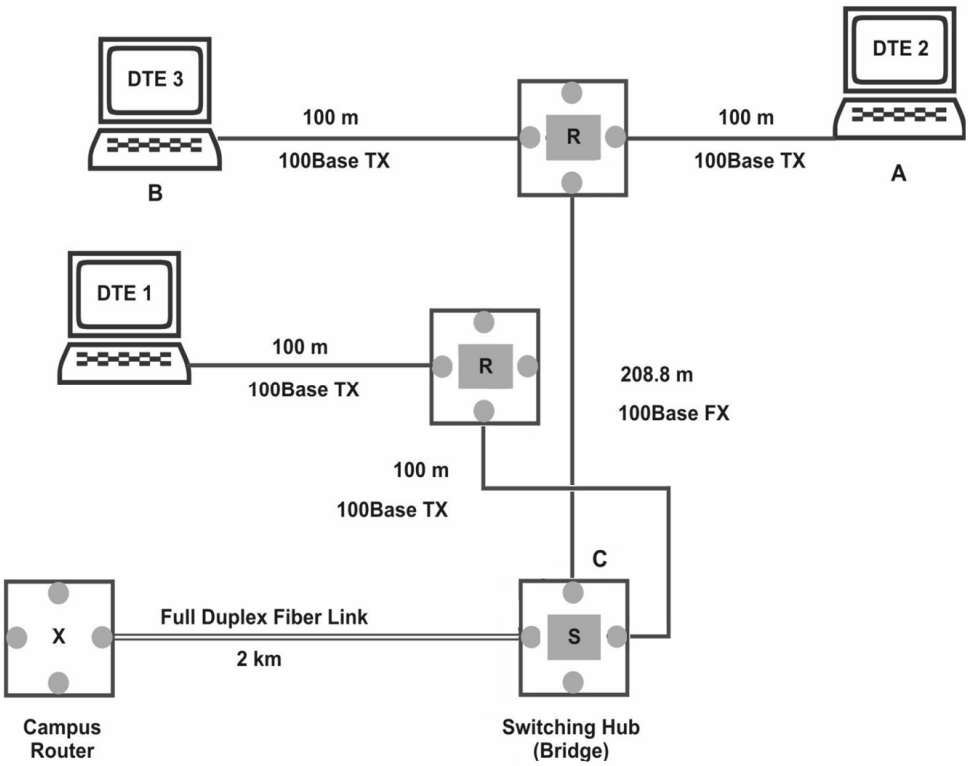


Figure 14.6
Mixed fiber and copper 100 Mbps configuration

A Class II repeater can be used to link the copper (TX) and fiber (FX) segments, since these segments both use the same encoding scheme.

14.6 Model 2 configuration guidelines for Fast Ethernet

Transmission system Model 2 for Fast Ethernet segments provides a set of calculations for verifying the signal-timing budget of more complex half-duplex Fast Ethernet LANs. These calculations are much simpler than the model 2 calculations used in the original 10 Mbps system, since the Fast Ethernet system uses only link segments.

The maximum diameter and the number of segments and repeaters in a half-duplex 100BaseT systems are limited by the round-trip signal timing required to ensure that the collision detect mechanism will work correctly. The Model 2 configuration calculations provide the information needed to verify the timing budget of a set of standard 100BaseT

segments and repeaters. This ensures that their combined signal delays fit within the timing budget required by the standard.

It may be noticed that these calculations appear to have a different round-trip timing budget than the timing budget provided in the 10 Mbps media system. This is because media segments in the Fast Ethernet system are based on different signaling systems than 10 Mbps Ethernet, and because the conversion of signals between the Ethernet interface and the media segments consumes a number of bit times.

It may also be noted that that there is no calculation for inter-frame gap shrinkage, unlike the one found in the 10 Mbps Model 2 calculations. That's because the maximum number of repeaters allowed in a Fast Ethernet system is limited, thus eliminating the risk of excessive inter-frame gap shrinkage.

14.6.1 Calculating round-trip delay time

Once the worst-case path has been found, the next step is to calculate the total round-trip delay. This can be accomplished by taking the sum of all the delay values for the individual segment in the path, plus the station delays and repeater delays. The calculation model in the standard provides a set of delay values measured in bit times, as shown in Table 14.5.

Component	Round-Trip Delay in Bit Times per m.	Maximum Round-Trip Delay in Bit Times per m.
Two TX/FX DTEs	N/A	100
Two T4 DTEs	N/A	138
One T4 & one TX/FX DTE (1)	N/A	127
Category 3 Cable	1.14	114 (100 m.)
Category 4 Cable	1.14	114 (100 m.)
Category 5 Cable	1.112	111.2 (100 m.)
Shielded Twisted Pair Cable	1.112	111.2 (100 m.)
Fiber Optic Cable	1.0	412 (412 m.)
Class I Repeater	N/A	140
Class II Repeater with all ports TX/FX	N/A	92
Class II Repeater with any T4 Port	N/A	67

Note (1): Worst Case values are used

Table 14.5
100BaseT component delays

It may be noted that the Round-Trip Delay in Bit Times per Meter only applies to the cable types in the table. The device types in the table (DTE, repeater) have only a maximum round-trip delay through each device listed.

To arrive at the round-trip delay value, multiply the length of the segment (in meters) times the round-trip delay in bit times per meter listed in the table for the segment type. This results in the round-trip delay in bit times for that segment. If the segment is at the maximum length one can use the maximum round-trip delay in bit times value listed in the table for that segment type. If not sure, of the segment length, one can also use the maximum length in your calculations just to be safe.

Once the segment delay values for each segment in the worst-case path are calculated, add the segment delay values together. One should also add the delay values for two stations (DTEs), and the delay for any repeaters in the path, to find the total path delay. The vendor may provide values for cable, station, and repeater timing, which one can use instead of the ones in the table.

To this total path delay value, add a safety margin of zero to four bit times, with four bit times of margin recommended in the standard. This helps account for unexpected delays, such as those caused by long patch cables between a wall jack in the office and the computer. If the result is less than or equal to 512 bit times, the path passes the test.

14.6.2 Calculating segment delay values

The segment delay value varies depending on the kind of segment used, and on the quality of cable in the segment if it is a copper segment. More accurate cable delay values may be provided by the manufacturer of the cable. If propagation delay of the cable being used is known, one can also look up the delay for that cable in Table 14.6 given below.

Speed relative to speed of light	Bit Time Per Meter		
	Nanoseconds per meter	100 Mbps Fast Ethernet	1000 Mbps Gigabit Ethernet
0.4	8.34	0.834	8.34
0.5	6.67	0.667	6.67
0.51	6.54	0.654	6.54
0.52	6.41	0.641	6.41
0.53	6.29	0.629	6.29
0.54	6.18	0.618	6.18
0.55	6.06	0.606	6.06
0.56	5.96	0.596	5.96
0.57	5.85	0.585	5.85
0.58	5.75	0.575	5.75
0.5852	5.70	0.57	5.7
0.59	5.65	0.565	5.65
0.60	5.56	0.556	5.56
0.61	5.47	0.547	5.47
0.62	5.38	0.538	5.38
0.63	5.29	0.529	5.29
0.64	5.21	0.521	5.21
0.65	5.13	0.513	5.13
0.654	5.10	0.51	5.10
0.66	5.05	0.505	5.05
0.666	5.01	0.501	5.01
0.67	4.98	0.498	4.98
0.68	4.91	0.491	4.91
0.69	4.83	0.483	4.83
0.70	4.77	0.477	4.77
0.80	4.17	0.417	4.17
0.90	3.71	0.371	3.71

Table 14.6
Conversion table for cable propagation times

Table 14.6 values are taken from the standard and provide a set of delay values in bit times per meter. These are listed in terms of the speed of signal propagation on the cable. The speed (propagation time) is provided as a percentage of the speed of light. This is also called the nominal velocity of propagation, or NVP, in vendor literature.

If one knows the NVP of the cable being used, then this table can provide the delay value in bit times per meter for that cable. A cable's total delay value can be calculated by multiplying the bit time/meter value by the length of the segment. The result of this calculation must be multiplied by two to get the total round-trip delay value for the segment. The only difference between 100 Mbps Fast Ethernet and 1000 Mbps Gigabit Ethernet in the above table is that the bit time in Fast Ethernet is ten times longer than the bit time in Gigabit Ethernet. For example, since the bit time is one nanosecond in Gigabit Ethernet, a propagation time of 8.34 nanoseconds per meter translates to 8.34 bit times.

14.6.3 Typical propagation values for cables

Typical propagation rates for Category 5 cable provided by two major vendors are given below. These values apply to both 100 Mbps Fast Ethernet and 1000 Mbps Gigabit Ethernet systems.

AT&T: Part No. 1061, Jacket: Non-Plenum, NVP= 70%
AT&T: Part No. 2061, Jacket: Plenum, NVP= 75%
Belden: Part No. 1583A, Jacket: Non-Plenum, NVP= 72%
Belden: Part No. 1585A, Jacket: Plenum, NVP= 75%

14.7 Model 1 configuration guidelines for Gigabit Ethernet

Transmission system Model 1 rules for half-duplex Gigabit Ethernet configuration are:

- The system is limited to a single repeater
- Segment lengths are limited to the lesser of 316 meters (1,036.7 feet) or the maximum transmission distance of the segment media type

The maximum length in terms of bit times for a single segment is 316 meters. However, any media signaling limitations, which reduce the maximum transmission distance of the link to below 316 meters, take precedence. Table 14.7 shows the maximum collision domain diameter for a Gigabit Ethernet system for the segment types shown. The maximum diameter of the collision domain is the longest distance between any two stations (DTEs) in the collision domain.

Configuration Configuration	Cat. 5 UTP	1000BASE CX	Fiber Optic 1000BASE -SX/LX	Cat.5 + Fiber Optic	1000BASE -CX and 1000BASE SX/LX
DTE-DTE Single Segment	100	25	316 (1)	N/A	N/A
One Repeater	200	50	220	210 (2)	220 (3)

Notes: (1) May be limited by the max. Transmission distance of the link
(2) Assumes 100 m. Of Cat. 5 unshielded twisted-pair (UTP) Cable and one fiber-optic link of 110 m.
(3) Assumes 25 m. Of 1000BASE-CX and one optical fiber link of 195 m.

Table 14.7
Model I maximum gigabit Ethernet collision domain in meters

The first row in table 14.7 shows the maximum lengths for a DTE-to-DTE (station-to-station) link. With no intervening repeater, the link may be up of a maximum of 100 m of copper, 25 m of 1000BaseCX cable, or 316 m of fiber optic cable. Some of the Gigabit Ethernet fiber optic links are limited to quite a bit less than 316 m due to signal transmission considerations. In those cases, one will not be able to reach the 316 m maximum allowed by the bit-timing budget of the system.

The row labeled *one repeater* provides the maximum collision domain diameter when using the single repeater allowed in a half-duplex Gigabit Ethernet system. This includes the case of all twisted-pair cable (200 m), all fiber optic cable (220 m) or a mix of fiber optic and copper cables.

14.8 Model 2 configuration guidelines for Gigabit Ethernet

Transmission system Model 2 for Gigabit Ethernet segments provides a set of calculations for verifying the signal-timing budget of more complex half-duplex Gigabit Ethernet LANs. These calculations are much simpler than the Model 2 calculations for either the 10 Mbps or 100 Mbps Ethernet systems, since Gigabit Ethernet only uses link segments and only allows one repeater. Therefore, the only calculation needed is the worst-case path delay value (PDV).

14.8.1 Calculating the path delay value

Once worst-case path has been determined, calculate the total round-trip delay value for the path, or PDV. The PDV is made up of the sum of segment delay values, repeater delay, DTE delays, and a safety margin.

The calculation model in the standard provides a set of delay values measured in bit times, as shown in Table 14.8. To calculate the round-trip delay value, multiply the length of the segment (in meters) times the round-trip delay in bit times per meter listed in the table for the segment type. This results in the round-trip delay in bit times for that segment.

Component	Round-trip Delay in Bit times per meter	Max. Round-trip in Bit Times
Two DTEs	N/A	864
Cat. 5 UTP Cable Segment	11.12	1112 (100 m)
Shielded Jumper Cable (CX)	10.10	253 (25 m)
Fiber Optic Cable Segment	10.10	1111 (110 m)
Repeater	N/A	976

Table 14.8
1000BaseT components delays

The result of this calculation is the round-trip delay in bit times for that segment. One can use the maximum round-trip delay in bit times value listed in the table for that segment type if the segment is at the maximum length. The max delay values can also be used if one is not sure of the segment length and want to use the maximum length in calculations just to be safe. To calculate cable delays, one can use the conversion values provided in right-hand column of Table 14.6.

To complete the PDV calculation, add the entire set of segment delay values together, along with the delay values for two stations (DTEs), and the delay for any repeaters in the path. Vendors may provide values for cable, station and repeater timing, which one can use instead of the ones in the tables provide here.

To this total path delay value, add a safety margin of from zero to 40 bit times, with 32 bit times of margin recommended in the standard. This helps account for any unexpected delays, such as those caused by extra long patch cords between a wall jack in the office and the computer. If the result is less than or equal to 4,096 bit times, the path passes the test.

14.9 Sample network configurations

A few sample network configurations will be undertaken to show how the configuration rules work in the real world. The 10 Mbps examples will be the most complex, since the 10 Mbps, system has the most complex set of segments and timing rules. After that, a single example for the 100 Mbps system will be discussed, since the configuration rules

are much simpler for Fast Ethernet. There is no need for a Gigabit Ethernet example, as the configuration rules are extremely simple, allowing for only a single repeater hub. In addition, all Gigabit Ethernet equipment being sold today only supports full-duplex mode, which means there are no half-duplex Gigabit Ethernet systems.

14.9.1 Simple 10 Mbps model 2 configurations

Figure 14.7 shows a network with three 10BaseFL segments connected to a fiber optic repeater. Two of the segments are 2 km (2,000 m) in length, and one is 1.5 km in length.

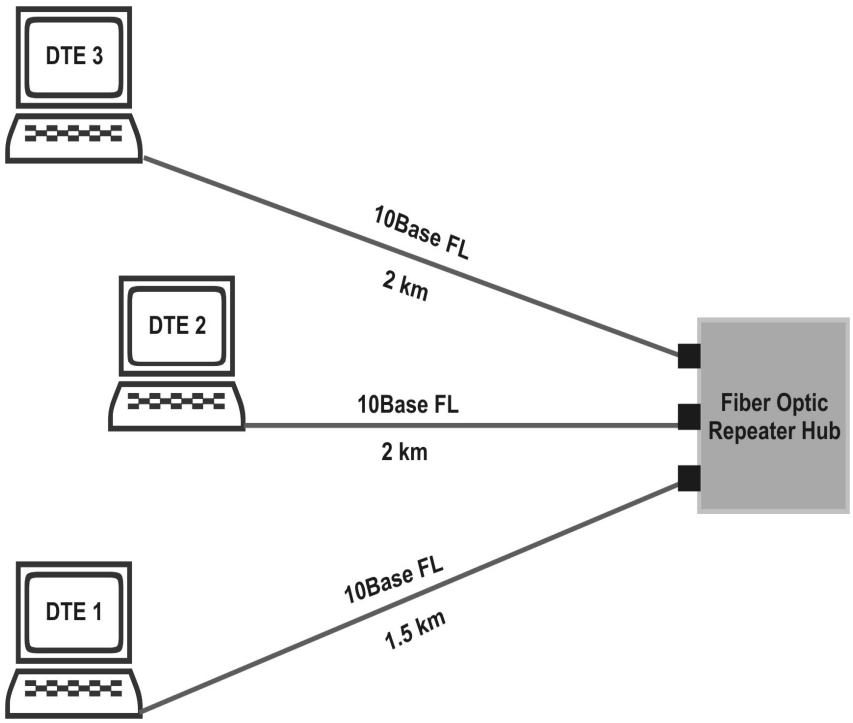


Figure 14.7
Simple 10 Mbps configuration example

This is a simple network, but its configuration is not described in the Model 1 configuration rules. The only way to verify its operation is to perform the Model 2 calculations. Figure 14.7 shows that that the worst-case delay path is between station 3 and station 2.

14.9.2 Round-trip delay

There are only two media segments in the worst-case path, and hence the network model for round-trip delay only has a left and right end segment. There are no middle segments to deal with. For the purposes of this example it shall be assumed that the fiber optic transceivers are connected directly to the stations and repeater. This eliminates the need to add extra bit times for transceiver cable length. Both segments in the worst-case path are the maximum allowable length. This means that using the 'max' values from Table 14.2 is the simplest way of calculating this.

According to the table, the max. left end segment delay value for a 2 km 10BaseFL link is 212.25 bit times. For the 2 km right end segment, the max. delay value is 356.5 bit times. Add them together, plus the five bit times margin recommended in the standard, and the total is: 573.75 bit times. This is less than the 575 maximum bit time budget

allowed for a 10 Mbps network, which means that the worst-case path is okay. All shorter paths will have smaller delay values; so all paths in this Ethernet system meet the requirements of the standard as far as round-trip timing is concerned. To complete the interframe gap calculation, one will need to compute the gap shrinkage in this network system.

14.9.3 Inter-frame gap shrinkage

Since there are only two segments, one only looks at a single transmitting end segment when calculating the inter-frame gap shrinkage. There are no middle segments to deal with, and the receive end segment does not count in the calculations for inter-frame gap. Since both segments are of the same media type, finding the worst-case value is easy. According to Table 14.3, the inter-frame gap value for the link segments is 10.5 bit times, and that becomes our total shrinkage value for this worst-case path. This is well under the 49 bit times of inter-frame shrinkage allowed for a 10 Mbps network.

As one can see, the example network meets both the round-trip delay requirements and the inter-frame shrinkage requirements, thus it qualifies as a valid network according to the Model 2 configuration method.

14.9.4 Complex 10 Mbps Model 2 configurations

The next example is more difficult, comprised of many different segment types, extra transceiver cables, etc. All these extra bits and pieces also make the example more complicated to explain, although the basic process of looking up the bit times and adding them together is still quite simple.

For this complex configuration example, refer back to figure 14.2 earlier in the chapter. This figure shows one possible maximum-length system using four repeaters and five segments. According to the Model 1 rule-based configuration method, it has already been seen that this network complies with the standards. To check that, one has to check this network again, this time using the calculation method provided for model 2.

First step is finding the worst-case path in the sample network. By examination, one can see that the path between stations 1 and 2 in figure 14.2 is the maximum delay path. It contains the largest number of segments and repeaters in the path between any two stations in the network. Next, one makes a network model out of the worst-case path. Start the process by arbitrarily designating the thin Ethernet end segment as the left end segment. That leaves three middle segments composed of a 10Base5 segment and two fiber optic link segments, and a right end segment comprised of a 10BaseT link segment.

Next, one has to calculate the segment delay value for the 10Base2 left end segment. This can be accomplished by adding the left end base value for 10Base2 coax (11.75) to the product of the round-trip delay times the length in meters ($185 \times 0.1026 = 18.981$) results in a total segment delay value of 30.731 for the thin coax segment. However, since 185 m is the maximum segment length allowed for 10Base2 segments, one can simply look up the max left hand segment value from table 14.2, which, not surprisingly, is 30.731. The 10Base2 thin Ethernet segment is shown attached directly to the DTE and the repeater, and there is no transceiver cable in use. Therefore, one does not have to add any excess AUI cable length timing to the value for this segment.

14.9.5 Calculating separate left end values

Since the left and right end segments in worst-case path are different media types, the standard notes that one needs to do the path delay calculations twice. First, calculate the total path delay using the 10Base2 segment as the left end segment and the 10BaseT segment as the right end. Then swap their places and make the calculation again, using the 10Base-T segment as the left end segment this time and the 10Base2 segment as the right end segment. The largest value that results from the two calculations is the value one must use in verifying the network.

14.9.6 AUI delay value

The segment delay values provided in the table include allowances for a transceiver cable (AUI) of up to two meters in length at each end of the segment. This allowance helps takes care of any timing delays that may occur due to wires inside the ports of a repeater.

Media systems with external transceivers connected with transceiver cables require that we account for the timing delay in these transceiver cables. One can find out how long the transceiver cables are, and use that length multiplied by the round-trip delay per meter to develop an extra transceiver cable delay time, which is then added to the total path delay calculation. If one is not sure how long, the transceiver cables are in your network, one can use the maximum delay shown for a transceiver cable, which is 4.88 for all segment locations, left end, middle, or right end.

14.9.7 Calculating middle segment values

In the worst-case path for the network in figure 14.2, there are three middle segments composed of a maximum length 10Base5 segment, and two 500 m long 10BaseFL fiber optic segments. By looking in table 14.2 under the Middle Segments column, one finds that the 10Base5 segment has a Max delay value of 89.8.

Note that the repeaters are connected to the 10Base5 segment with transceiver cables and outboard MAUs. That means one needs to add the delay for two transceiver cables. Let's assume that one does not know how long the transceiver cables are. Therefore, one has to use the value for two maximum-length transceiver cables in the segment, one at each connection to a repeater. That gives a transceiver cable delay of 9.76 to add to the total path delay.

One can calculate the segment delay value for the 10BaseFL middle segments by multiplying the 500-meter length of each segment by the RT Delay/meter value, which is 0.1, giving a result of 50. Add 50 to the middle segment base value for a 10Base-FL segment, which is 33.5, for a total segment delay of 83.5.

Although it's not shown in Figure 14.2, fiber optic links often use outboard fiber optic transceivers and transceiver cables to make a connection to a station. Just to make things a little harder, let it be assumed that one uses two transceiver cables, each being 25 m in length, to make a connection from the repeaters to outboard fiber optic transceivers on the 10Base-FL segments. That gives a total of 50 m of transceiver cable on each 10BaseFL segment. Since now there are two such middle segments, one can represent the total transceiver cable length for both segments by adding 9.76 extra bit times to the total path delay.

14.9.8 Completing the round-trip timing calculation

Here our calculations started with the 10Base2 segment assigned to the left end segment, which leaves us with a 10BaseT right end segment. This segment is 100 m long, which is the length provided in the 'Max' column for a 10Base-T segment. Depending on the cable

quality, a 10BaseT segment can be longer than 100 m, but we'll assume that the link in our example is 100 m. That makes the Max value for the 10BaseT right end segment 176.3. Adding all the segment delay values together, one gets the result shown in table 14.9.

Link	Media	Bit-Time Delay
Left End	10BASE-T	30.731
Mid-Segment	10BASE5	89.8
Mid-Segment	10BASE-FL	83.5
Mid-Segment	10BASE-FL	83.5
Right End	10Base-T	176.3
Excess Length		
AUI	Quan. Four	19.52
Total Path Delay		483.351

Table 14.9
Round-trip path delay with 10Base2 left end segments

To complete the process, one needs to perform a second set of calculations with the left and right segments swapped. In this case, the left end becomes a maximum length 10BaseT segment, with a value of 26.55, and the right end becomes a maximum length 10Base2 segments with a value of 188.48. Note that the excess length AUI values do not change. As shown in Table 14.2, the bit time values for AUI cables are the same no matter where the cables are used. Adding the bit time values again, one gets the following result in Table 14.10.

Link	Media	Bit-Time Delay
Left End	10BASE-T	26.55
Mid-Segment	10BASE5	89.8
Mid-Segment	10BASE-FL	83.5
Mid-Segment	10BASE-FL	83.5
Right End	10Base-2	188.48
Excess Length		
AUI	Quan. Four	19.52
Total Path Delay		491.35

Table 14.10
Round-trip path delay with 10BaseT left end segments

The second set of calculations shown in table 14.10 produced a larger value than the total from Table 14.9. According to the standard, one must use this value for the worst-case round-trip delay for this Ethernet. The standard also recommends adding a margin of five bit times to form the total path delay value. One is allowed to add anywhere from zero to five bits margin, but five bit times is recommended.

Adding five bit times for margin brings us up to a total delay value of 496.35 bit times, which is less than the maximum of 575 bit times allowed by the standard. Therefore, the complex network is qualified in terms of the worst-case round-trip timing delay. All shorter paths will have smaller delay values, which means that all paths in the Ethernet system shown in figure 14.2 meet the requirements of the standard as far as round-trip timing is concerned.

14.9.9 Inter-frame gap shrinkage

Evaluation of the complex network example shown in figure 14.2 by calculating the worst-case interframe gap shrinkage for that network is now considered. This is done by evaluating the same worst-case path we used in the path delay calculations. However, for the purposes of calculating gap shrinkage evaluate only the transmitting and mid-segments.

Once again one starts by applying a network model to the worst-case path, in this case the network model for inter-frame gap shrinkage. To calculate inter-frame gap shrinkage, the transmitting segment should be assigned the end segment in the worst-case path of your network system that has the largest shrinkage value. As shown in table 14.3, the coax media segment has the largest value, so for the purposes of evaluating our sample network we will assign the 10Base2 thin coax segment to the role of transmitting end segment. That leaves us with middle segments consisting of one coax and two link segments, and a 10Base-T receive end segment which is simply ignored. The totals are shown below:

PVV for transmitting End Coax	=	16
PVV for Mid-Segment Coax	=	11
PVV for Mid-Segment Link	=	8
PVV for Mid-Segment Link	=	8
Total PVV	**=**	**43**

It can be seen that, the total path variability value for our sample network equals 43. This is less than the 49-bit time maximum allowed in the standard, which means that this network meets the requirements for inter-frame gap shrinkage.

14.9.10 100 Mbps model 2 configuration

For this example, refer back to figure 14.5, which shows one possible maximum length network. As we've seen, the Model 1 rule-based configuration method shows that this system is okay. To check that, we'll evaluate the same system using the calculation method provided in Model 2.

14.9.11 Worst-case path

In the sample network, the two longest paths are between Station 1 and Station 2, and between Station 1 and the switching hub. Signals from Station 1 must go through two repeaters and two 100 m segments, as well as a 5 m inter-repeater segment to reach either Station 2 or the switching hub. As far as the configuration guidelines are concerned, the switching hub is considered as another station.

Both of these paths in the network include the same segment lengths and number of repeaters, so we will evaluate one of them as the worst-case path. Let's assume that all three segments are 100Base-TX segments, based on Category 5 cables. By looking up the Max Delay value in Table 14.5 for a Category 5 segment, we find 111.2 bit times.

The delay of a 5 m inter-repeater segment can be found by multiplying the round-trip Delay per Meter for Category 5 cable (1.112) times the length of the segment in meters (5). This results in 5.56 bit times for the round-trip delay on that segment. Now that we know the segment round-trip delay values, we can complete the evaluation by following the steps for calculating the total round-trip delay for the worst-case path.

To calculate the total round-trip delay, we use the delay times for stations and repeaters found in table 14.5. As shown in below, the total round-trip path delay value for the sample network is 511.96 bit times when using Category 5 cable. This is less than the maximum of 512 bit times, which means that the network passes the test for round-trip delay.

Delay for two TX DTEs	=	100
Delay for 100 m. Cat. 5 segment	=	111.2
Delay for 100 m. Cat. 5 segment	=	111.2
Delay for 5 meter Cat. 5 segment	=	5.56
Delay for Class II repeater	=	92
Delay for Class II repeater	=	92
Total Round-Trip Delay	**=**	**511.96 bits**

It may be noted that there is no margin of up to 4 bit times provided in this calculation. There are no spare bit times to use for margin, because the bit time values shown in Table 14.5 are all worst-case maximums. This table provides worst-case values that you can use if you don't know what the actual cable bit times, repeater timing, or station-timing values are.

For a more realistic look, let's see what happens if we work this example again, using actual cable specifications provided by a vendor. Assume that the Category 5 cable is AT&T type 1061 cable, a non-plenum cable that has an NVP of 70 percent as shown below. If we look up that speed in table 14.6, we find that a cable with a speed of 0.7 is rated at 0.477 bit times per meter. The round-trip bit time will be twice that, or 0.954 bit times. Therefore, timing for 100 m will be 95.4 bit times, and for 5m it will be 4.77 bit times. How things add up using these different cable values is shown below:

Delay for Two TX DTEs	=	100
Delay for 100 m. Cat. 5 Segment	=	95.4
Delay for 100 m. Cat. 5 segment	=	95.4
Delay for 5 m. Cat. 5 Segment	=	4.77
Delay for Class II repeater	=	92
Delay for Class II repeater	=	92
Total Delay	**=**	**483.57 bits**

When real-world cable values are used instead of the worst-case default values in table 7.5, there is enough timing left to provide for 4 bit times of margin. This meets the goal of 512 bit times, with bit times to spare.

14.9.12 Working with bit time values

Some vendors note that their repeater delay values are smaller than the values listed in Table 14.5, which will make it easier to meet the 512-bit time maximum. While these extra bit times could theoretically be used to provide an inter-repeater segment longer than five meters, this approach could lead to problems.

While providing a longer inter-repeater link might appear to be a useful feature, one should consider what would happen if that vendor's repeater failed and had to be replaced with another vendor's repeater whose delay time was larger. If that were to occur, then the worst-case path in your network might end up with excessive delays due to the bit times consumed by the longer inter-repeater segment you had implemented. One can

avoid this problem by designing the network conservatively and not pushing things to the edge of the timing budget.

One can use more than one Class I or two Class II repeaters in a given collision domain. This can be done if the segment lengths are kept short enough to provide the extra bit time budget required by the repeaters. However, the majority of network installations are based on building cabling systems with 100 m segment lengths (typically implemented as 90 m 'in the walls' and 10 m for patch cables). A network design with so many repeaters that the segments must be kept very short to meet the timing specifications is not going to be useful in most situations.